



Panchip Microelectronics Co., Ltd.

PAN3029 plus PA reference design and test data

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Document Description

None

Revision History

Version	Revision Time	Description
V1.0	2024.02	Initial version created

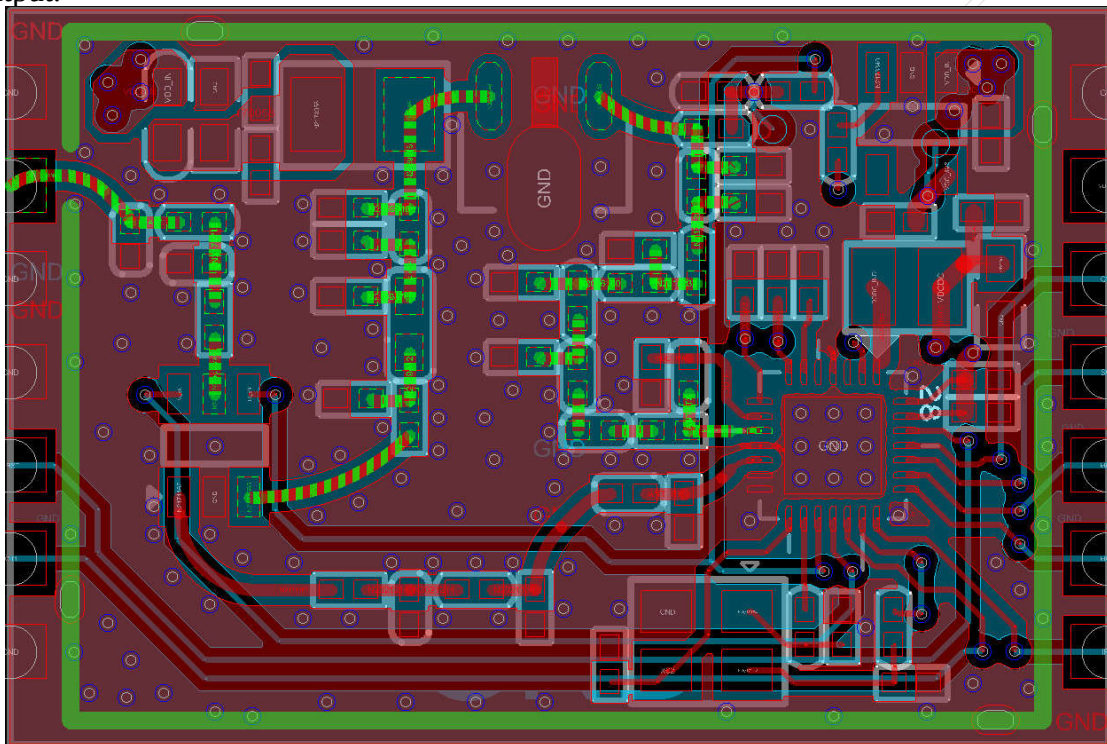
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1 PCB Reference Layout

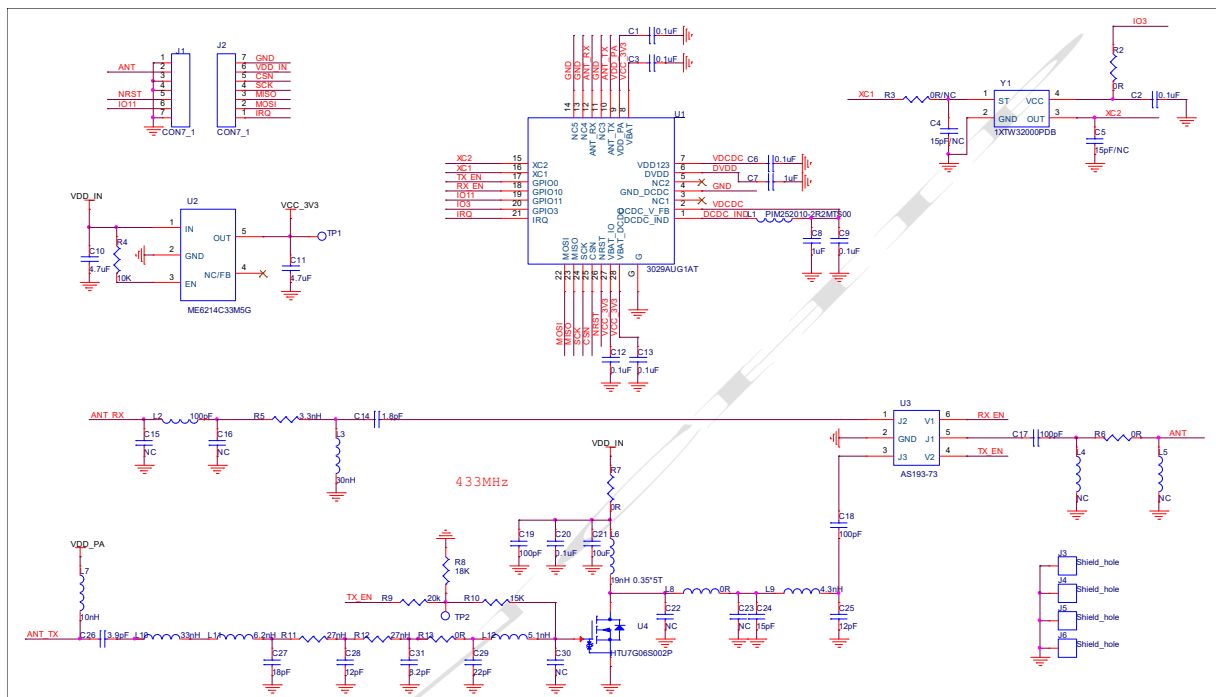
1.1 PCB reference layout for all frequency bands

The reference design for all frequency bands is debugged based on the following PCB layout. The green highlighted part is the RF transmission circuit. The VDD_IN power supply is 5V, and the PA efficiency is calculated based on the PA power consumption and the output power of the switch output.



2 433MHz原理图和性能参数

2.1 433MHz 原理图

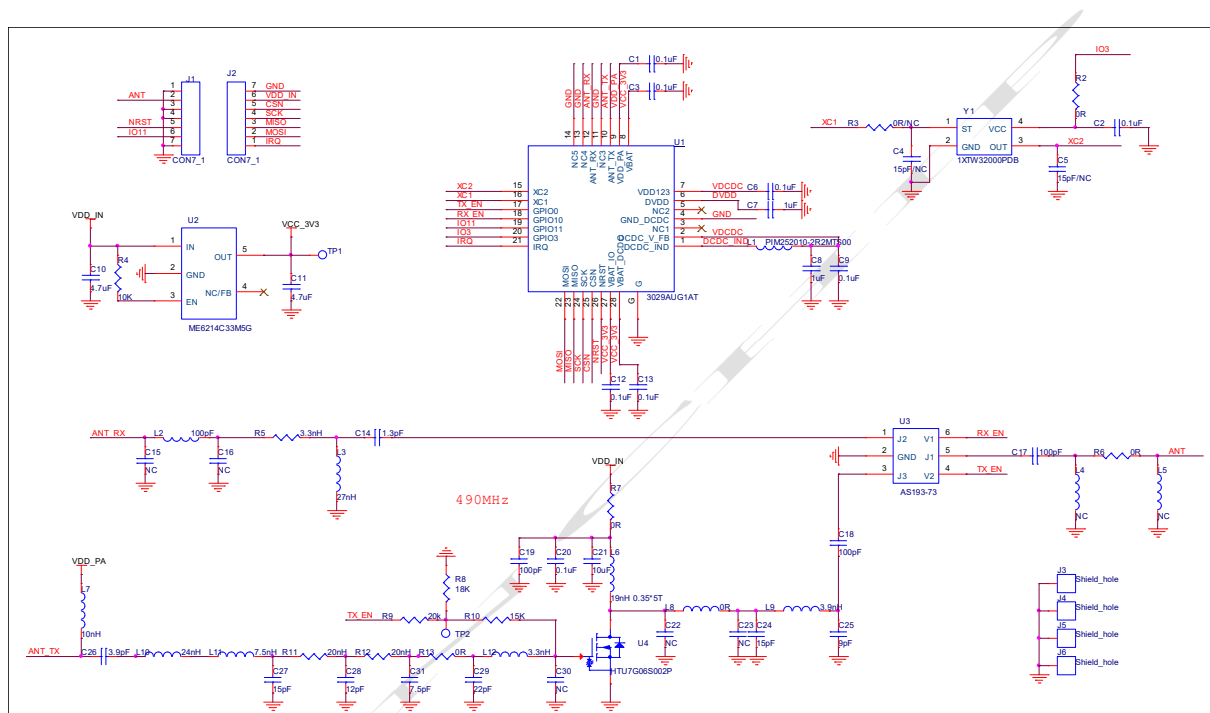


2.1 433MHz 射频性能

频率/MHz	模组输出功率/dBm	模组功耗/mA	PA 效率
410	34.139	1125.8	55.01%
415	34.089	1080.9	55.47%
420	34.009	1050.3	56.01%
425	33.913	1029.7	56.55%
430	33.818	1015.1	56.97%
435	33.722	1005.3	57.29%
440	33.625	1000.7	57.58%
445	33.538	1002.7	57.81%
450	33.472	1010.1	57.81%
455	33.409	1013.8	57.46%
460	33.295	995.3	56.94%
465	32.769	946	54.56%
470	31.883	823	50.37%

3 490MHz原理图和性能参数

3.1 490MHz 原理图

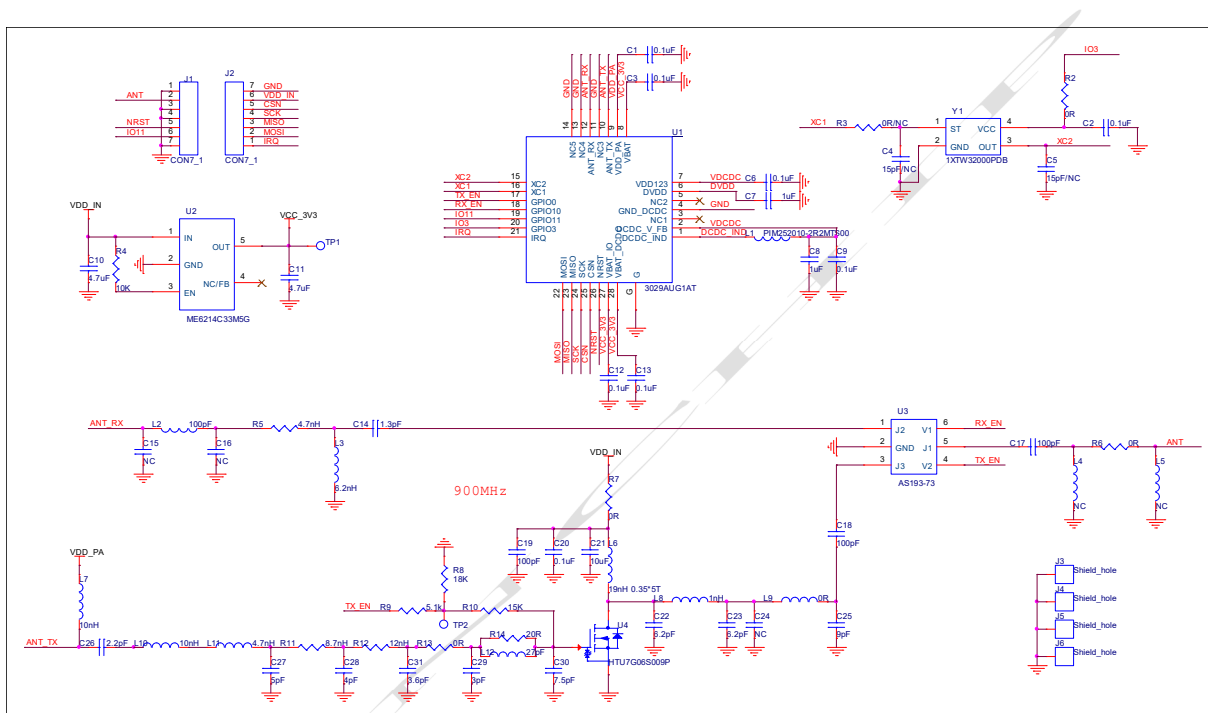


3.2 490MHz 射频性能

频率/MHz	模组输出功率/dBm	模组功耗/mA	PA 效率
465	33.685	1142.7	40.89%
470	33.675	1103	42.26%
475	33.634	1040	44.40%
480	33.575	981.3	46.42%
485	33.491	934.4	47.82%
490	33.408	899.2	48.75%
495	33.338	873.4	49.39%
500	33.264	854	49.66%
505	33.189	839.3	49.66%
510	33.09	827.9	49.21%
515	32.975	818.7	48.46%

4 868和915MHz原理图和性能

4.1 868 和 915MHz 原理图



4.2 868 和 915MHz 射频性能

频率/MHz	模组输出功率/dBm	模组功耗/mA	PA 效率
855	33.563	1142.037	39.78%
860	33.461	1125.565	39.42%
865	33.34	1111.529	38.82%
870	33.217	1100.609	38.12%
875	33.151	1092.782	37.81%
900	33.207	1051.838	39.79%
905	33.22	1042.113	40.28%
910	33.264	1033.472	41.03%
915	33.295	1026.988	41.59%
920	33.378	1022.307	42.58%
925	33.432	1017.869	43.31%
930	33.49	1011.492	44.16%

935	33.503	1000.831	44.77%
940	33.509	984.559	45.57%

